

Strict Positivity and Simplicity of the Ground State for a Functional-Difference Operator

Candidate full solution packet for arXiv:2109.05465

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Status. Candidate full affirmative answer; likely valid, pending human review. Under the hypothesis in the source problem, the lowest eigenvalue is simple and its eigenfunction has a continuous representative that is strictly positive at every real point.

1 Source question

Let $P = -i d/dx$, $b > 0$, and

$$W_0(b) = 2 \cosh(bP), \quad W_V(b) = W_0(b) - V$$

on $L^2(\mathbb{R})$, where $0 \leq V \in L^1(\mathbb{R})$ and $W_V(b)$ is defined by its Friedrichs form. Laptev and Schimmer ask after Remark 4.1 of *A Sharp Lieb–Thirring Inequality for Functional Difference Operators*, arXiv:2109.05465 [1]:

Assume that $\sigma_d(W_V(b)) \subset [-2, 2)$. Is it true that the eigenfunction corresponding to the lowest eigenvalue could be chosen strictly positive?

is positive assuming that the coupling constant c is small enough satisfying the inequality $c/(2\pi b) \leq 1$ and thus $\omega_c \in [0, \pi)$. Note that if $c/(2\pi b) = 1$ then $\omega_c = 0$ and

$$\psi_c(x) = \frac{\pi \psi(0)x}{b \sinh\left(\frac{\pi}{b}x\right)} > 0.$$

However, if the coupling constant $c > 2\pi b$ then $\omega_c \in i(0, \infty)$ and hence

$$\psi_c(x) = \frac{c\psi(0)}{2b \sinh|\omega_c|} \frac{\sin\left(\frac{|\omega_c|}{b}x\right)}{\sinh\left(\frac{\pi}{b}x\right)}$$

is an oscillating function and in particular has an infinite number of zeros. This contradicts a possible conjecture that the eigenfunction for the lowest eigenvalue is strictly positive.

Open problem. Assume that the discrete spectrum $\sigma_d(W_V(b))$ of the operator $W_V(b)$ satisfies the property $\sigma_d(W_V(b)) \subset [-2, 2)$. Is it true that the eigenfunction corresponding to the lowest eigenvalue could be chosen strictly positive?

Figure 1: The exact question and its motivating rank-one calculation on source PDF page 9.

We give an affirmative answer and also prove simplicity. The case $V = 0$ has no discrete eigenvalue and is vacuous, so below V is assumed nonzero.

2 Proof intuition

The obstruction seen by the source below energy -2 is precisely the loss of positivity of the free Green function. On the entire interval $[-2, 2)$, including the endpoint -2 , that Green function is instead strictly positive. Consequently the Birman–Schwinger operator at the lowest eigenvalue is a compact positivity-improving integral operator on the support of V .

The only remaining point is to know which of its eigenvalues equals one. Birman–Schwinger counting says that an eigenvalue above one would produce an eigenvalue of $W_V(b)$ strictly below the alleged ground state. Thus one is the spectral radius. A short Perron argument makes its eigenspace one-dimensional with a strictly positive eigenvector. Applying the free resolvent to that vector reconstructs an L^2 eigenfunction which is a convolution of two nonnegative nonzero functions, one of them everywhere strictly positive.

3 Statement

Theorem 1. *Let $b > 0$ and let $0 \leq V \in L^1(\mathbb{R})$ be nonzero. Suppose*

$$\sigma_d(W_V(b)) \subset [-2, 2).$$

Then $W_V(b)$ has a lowest discrete eigenvalue λ_0 . This eigenvalue is simple, and its eigenspace is spanned by a function having a continuous representative ψ_0 such that

$$\psi_0(x) > 0 \quad (x \in \mathbb{R}).$$

In particular, the source question has an affirmative answer.

4 The positive free resolvent

The Fourier symbol of $W_0(b)$ is $2 \cosh(2\pi b k)$. For $\lambda < 2$ write $R_\lambda = (W_0(b) - \lambda)^{-1}$.

Lemma 2. *If $-2 \leq \lambda < 2$, then R_λ is convolution with a real, even function $G_\lambda \in C_0(\mathbb{R}) \cap L^2(\mathbb{R})$ which is strictly positive at every point.*

Proof. For $-2 < \lambda < 2$, write $\lambda = -2 \cos \omega$ with $0 < \omega < \pi$. The source computes

$$G_\lambda(t) = \frac{1}{2b \sin \omega} \frac{\sinh(\omega t/b)}{\sinh(\pi t/b)}, \quad G_\lambda(0) = \frac{1}{2\pi b} \frac{\omega}{\sin \omega}. \quad (1)$$

For $t \neq 0$, the two hyperbolic sines have the same sign. Thus $G_\lambda(t) > 0$, and the displayed diagonal value is positive. The quotient is continuous at zero and decays exponentially at infinity because $\omega < \pi$, proving the asserted regularity and integrability.

At $\lambda = -2$, passage to the limit $\omega \downarrow 0$ gives

$$G_{-2}(t) = \frac{t}{2b^2 \sinh(\pi t/b)}, \quad G_{-2}(0) = \frac{1}{2\pi b}. \quad (2)$$

This function has the same properties and is strictly positive everywhere. $\square$

Put $S = \{x : V(x) > 0\}$, modulo null sets, and use $L^2(S)$ as the natural nonzero support space. Define

$$K_\lambda = V^{1/2} R_\lambda V^{1/2}.$$

Lemma 3. For $-2 \leq \lambda < 2$, K_λ is a nonzero compact self-adjoint nonnegative operator on $L^2(S)$. It is positivity improving: if $0 \neq f \in L^2(S)$ and $f \geq 0$, then $K_\lambda f > 0$ almost everywhere on S .

Proof. Its integral kernel is

$$k_\lambda(x, y) = V(x)^{1/2} G_\lambda(x - y) V(y)^{1/2}.$$

Lemma 2 makes G_λ bounded, so

$$\|K_\lambda\|_{\text{HS}}^2 \leq \|G_\lambda\|_\infty^2 \|V\|_1^2 < \infty.$$

Thus K_λ is compact. It is self-adjoint because G_λ is real and even, and it is nonnegative in the Hilbert-space sense because $R_\lambda \geq 0$. Finally, $k_\lambda(x, y) > 0$ for almost every $(x, y) \in S \times S$. Integrating it against a nonzero nonnegative f proves positivity improvement and also nontriviality. $\square$

We record the elementary Perron fact needed below, including its proof so that no irreducibility theorem is hidden in the argument.

Lemma 4. Let K be a nonzero compact self-adjoint nonnegative integral operator on a real or complex L^2 -space. If its kernel is strictly positive almost everywhere, then $r = \|K\|$ is a simple eigenvalue and it has an eigenvector $\phi > 0$ almost everywhere.

Proof. Compactness and nonnegativity show that $r > 0$ is an eigenvalue. Since the kernel is real, choose a nonzero real eigenvector u . Pointwise kernel positivity gives

$$\langle |u|, K|u| \rangle \geq \langle u, Ku \rangle = r \|u\|_2^2.$$

The reverse inequality follows from $K \leq rI$. Hence equality holds and $|u|$ is again an r -eigenvector. Positivity improvement gives $K|u| > 0$ almost everywhere, so $|u| > 0$ almost everywhere.

If the real r -eigenspace had dimension at least two, it would contain a nonzero w orthogonal to the positive eigenvector. Both w_+ and w_- would then be nonzero. Strict positivity of the kernel gives

$$\langle |w|, K|w| \rangle - \langle w, Kw \rangle = 4 \langle w_+, Kw_- \rangle > 0,$$

contradicting $\langle |w|, K|w| \rangle \leq r \|w\|_2^2 = \langle w, Kw \rangle$. Thus the real eigenspace is one-dimensional, and the same follows for its complexification. $\square$

5 Birman–Schwinger counting at the ground state

We use the standard Birman–Schwinger principle in the same form setting as the source paper. For every $\lambda < 2$,

$$\#\{\text{eigenvalues of } W_V(b) \text{ below } \lambda\} = \#\{\text{eigenvalues of } K_\lambda \text{ above } 1\}, \quad (3)$$

with multiplicity. Moreover,

$$\ker(W_V(b) - \lambda) \longleftrightarrow \ker(K_\lambda - I) \quad (4)$$

is a linear isomorphism, initially given from left to right by $\psi \mapsto V^{1/2}\psi$.

For completeness, these statements follow by writing the form of $W_V(b) - \lambda$ through $(W_0(b) - \lambda)^{1/2}$ and applying the min–max principle to $I - R_\lambda^{1/2} V R_\lambda^{1/2}$. The latter operator and K_λ have the same nonzero eigenvalues. At eigenvalue one, the inverse map in (4) is

$$\phi \mapsto R_\lambda V^{1/2} \phi. \quad (5)$$

This is also the Birman–Schwinger principle invoked in Section 3.1 of [1].

Proof of Theorem 1. Because $V \geq 0$ is nonzero, Theorem 1.3 of [1] gives at least one eigenvalue below 2. The source also establishes that the essential spectrum is $[2, \infty)$ and that the spectrum below 2 consists of isolated finite-multiplicity eigenvalues. Such eigenvalues can accumulate only at 2, so the stated inclusion in $[-2, 2)$ gives a lowest one, denoted λ_0 .

The correspondence (4) says that 1 is an eigenvalue of K_{λ_0} . Since no eigenvalue of $W_V(b)$ lies below λ_0 , (3) says that K_{λ_0} has no eigenvalue strictly above 1. Lemma 3 makes this a nonnegative compact operator, and hence

$$\|K_{\lambda_0}\| = 1.$$

Lemmas 3 and 4 provide a unique (up to scalar) eigenvector $\phi \in L^2(S)$ for eigenvalue one, normalized so that $\phi > 0$ almost everywhere on S .

Set $h = V^{1/2}\phi$, extended by zero off S . Since $V^{1/2}\phi \in L^2(S)$, Cauchy–Schwarz gives $h \in L^1(\mathbb{R})$, and h is nonnegative and nonzero. Define

$$\psi = R_{\lambda_0}h = G_{\lambda_0} * h.$$

Lemma 2 and Young’s inequality give $\psi \in L^2(\mathbb{R})$. The Birman–Schwinger equation yields

$$V^{1/2}\psi = K_{\lambda_0}\phi = \phi, \quad V\psi = V^{1/2}\phi = h.$$

Thus (5), or equivalently the quadratic-form identity, shows that $W_V(b)\psi = \lambda_0\psi$.

There is no domain gap in this reconstruction. Indeed $\widehat{h}$ is bounded and

$$\int_{\mathbb{R}} \frac{2 \cosh(2\pi bk)}{(2 \cosh(2\pi bk) - \lambda_0)^2} |\widehat{h}(k)|^2 dk < \infty,$$

so ψ belongs to the form domain of $W_0(b)$. The equality $V\psi = h \in L^1$ then gives the eigenvalue identity first in the form sense; the representation theorem places ψ in the operator domain.

Finally, $G_{\lambda_0} \in C_0(\mathbb{R})$ and $h \in L^1(\mathbb{R})$, so the convolution is continuous. For every $x \in \mathbb{R}$,

$$\psi(x) = \int_{\mathbb{R}} G_{\lambda_0}(x - y)h(y) dy > 0,$$

because the Green function is everywhere strictly positive and h is nonnegative and nonzero. Simplicity of the eigenvalue one in Lemma 4, together with the eigenspace isomorphism (4), proves simplicity of λ_0 . $\square$

6 Verification, scope, and novelty

The proof covers the endpoint $\lambda_0 = -2$ explicitly through (2); no limiting eigenfunction argument is used. It uses only $V \in L^1$, not boundedness or strict positivity of V . Positivity is proved on all of $\mathbb{R}$, not merely almost everywhere, for the continuous convolution representative.

Bounded local-index, parsed-arXiv, exact-phrase, title/author, DOI/citation, and author-page searches were run on 11 August 2026. They found the source and adjacent 2025 nonselfadjoint work, but no later paper stating this answer. Novelty confidence is moderate; this is not a priority claim.

Human-review recommendation. Check form Birman–Schwinger counting for $V \in L^1$, the endpoint kernel (2), and reconstruction. The Perron argument is self-contained; no unproved dependency remains.

Reference

- [1] A. Laptev and L. Schimmer, *A Sharp Lieb–Thirring Inequality for Functional Difference Operators*, SIGMA **17** (2021), 105, 10 pp.; arXiv:2109.05465; doi:10.3842/SIGMA.2021.105.